QR CODE **WATCH THE VIDEO BEFORE YOU SOLVE** YouTube • Lesson 1

RULE · Multi-step equation (ko'p qadamli tenglama) – solving order

Work in order: clear fractions → expand parentheses → collect the variable on one side → combine like terms.

$$\frac{2}{3}(x+3) = x - 1 \qquad \leftarrow \text{1. clear the fraction } (\times 3)$$

$$\downarrow$$

$$2(x+3) = 3(x-1) \qquad \leftarrow \text{2. expand parentheses}$$

$$\downarrow$$

$$2x + 6 = 3x - 3 \qquad \leftarrow \text{3. collect the variable}$$

$$\downarrow$$

$$6 + 3 = 3x - 2x \qquad \leftarrow \text{4. combine like terms}$$

$$\downarrow$$

$$x = 9$$

WARNING · Always check what's being asked

Most common SAT error: solving correctly for x, then answering with x – when the question asks for a different expression. Re-read the question after you solve.

Question: $3x + 5 = 20$. Find $6x + 10$.

X Wrong: $x = 5$, answer "5".

Right:

$x = 5 \rightarrow 6(5) + 10 = 40$.

Answer: 40, not x!

RULE · Sign-flip rule (ishora almashish qoidasi)

Operation	Sign flips?
Add or subtract	No
Multiply / divide by a positive	No
Multiply / divide by a negative	**Yes – flips**

$-3x + 5 \leq 20 \;\Rightarrow\; -3x \leq 15$

(dividing by -3 – sign flips) $\Rightarrow\; x \geq -5$

RULES SHEET – 2/2

RULE · Graphing the solution

Sign	Circle	Meaning
$<$ or $>$	Open circle	Boundary excluded
$\leq$ or $\geq$	Closed circle	Boundary included

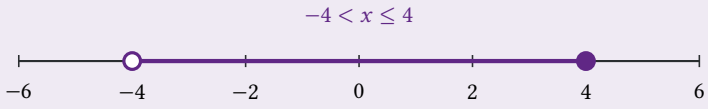
$x > 3$ **(open)**

$x \leq 4$ **(closed)**

RULE · Compound inequality (qo'shma tengsizlik)

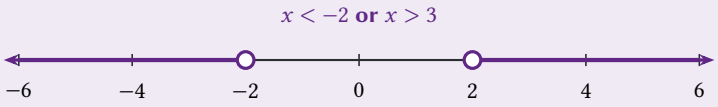
"And" type – same operation on all three parts:
$-3 < x + 1 \leq 5 \;\Rightarrow\; -4 < x \leq 4$

$-4 < x \leq 4$



"Or" type – solve each part separately:
$2x + 1 < -3$ or $2x + 1 > 7 \;\Rightarrow\; x < -2$ or $x > 3$

$x < -2$ **or** $x > 3$



WARNING: the operation must apply to **all three parts** of a three-part inequality – not just one.

RULE · Notation note: interval notation (oraliq belgilash)

Traditional Uzbek textbooks use a semicolon; SAT and English-language sources use a comma – same meaning:

Traditional: $(2;\ 5]$ SAT / English: $(2,\ 5]$

Always use a comma in your own work – matches the SAT format.